# MORISH ALFONSO R. MACAYAN

Computer Science Student | Artificial Intelligence Track

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## PROFILE

BS Computer Science student on the Artificial Intelligence track with experience in machine learning research, LLM tool integration, and web development. Builds reproducible workflows with baseline comparisons, automated tests, and documented limitations. Interested in applied ML and developer tools.

## EDUCATION

### Mapúa University

2024 - Present

BS Computer Science - Artificial Intelligence Track

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, PHP, SQL, Java, C/C++
**ML & data:** scikit-learn, pandas, NumPy, model evaluation, feature engineering, LLM tool use
**Web & databases:** HTML/CSS, Flask, Tailwind CSS, REST APIs, MySQL
**Tools:** Git, GitHub, Jupyter Notebook, JavaFX, Maven, web deployment

## EXPERIENCE

### Machine Learning Intern | FlyRank

2026

- Built an end-to-end ML workflow using anonymized search-performance data to prioritize content refresh opportunities.
- Performed preprocessing, feature engineering, model evaluation, and baseline comparisons; documented methods and ranked recommendations in a reproducible research report.
- Published interactive visualizations and research outputs through GitHub Pages using public-safe reporting practices.

### Front-End Developer | Self-Employed

2025

- Designed and developed four client websites: Solace, Celesté, Thrift Nest, and Furever Haven using HTML, CSS, and JavaScript.
- Managed client consultation, interface design, implementation, and deployment independently.

## PERSONAL PROJECTS

### LeakageLens | Dataset Leakage Auditor

Python, scikit-learn, JavaScript | Source code

- Built a local CSV auditor for exact train/test overlap, shared entities, time-order issues, and potentially target-derived features, with a CLI and browser interface.
- Trained a random-forest column reviewer and benchmarked it on 760 synthetic profiles from 40 source datasets; included dataset-level splits, rule baselines, transfer checks, and 33 automated tests.

### TutorialCheck | pandas Compatibility Reviewer

Python, pandas, scikit-learn, JavaScript | Source code

- Combined AST-based checks with TF-IDF and logistic regression to flag outdated pandas patterns and suggest migrations without executing user-submitted code.
- Generated 414 execution-labeled fixtures, verified 198 repaired examples, and compared the classifier with static rules on an independently authored challenge set.

## PERSONAL PROJECTS / CONTINUED

### TravelEase | Trip Planning & Support Application

PHP, MySQL, HTML/CSS, JavaScript, Groq API | Source code

- Developed trip CRUD, day-by-day itineraries, and estimated-versus-actual budget tracking with customer, staff, and admin roles.
- Implemented Google sign-in with server-side token verification, password hashing, CSRF protection, and ownership-scoped database queries.
- Integrated an AI support assistant with rule-based escalation and staff messaging for requests requiring human support.

### Data Scout Agent | Data Analysis Assistant

Python, pandas, Groq API, LLM tool calling | Source code

- Created a command-line assistant that answers natural-language questions about content performance by generating and executing pandas queries.
- Exposed executed queries and results, required numerical answers to be grounded in tool output, and capped tool-use turns to bound the agent loop.

### Multilayer Perceptron from Scratch

Python, NumPy, Jupyter Notebook

- Implemented a two-layer MLP with forward propagation, backpropagation, configurable activation functions, and mini-batch gradient descent; evaluated training and validation metrics across hyperparameter settings.

## ACADEMIC PROJECTS

### ALX Desserts | Team Lead & Full-Stack Developer

Python, Flask, JavaScript, Tailwind CSS | Source code

- Developed a web-based POS and inventory system with GCash QR payment, sales reporting, and receipt generation.

### HotelEase | Lead Programmer 2

Java, JavaFX, Maven | Source code

- Contributed application logic, base templates, GUI, and interactivity to a hotel management and booking system with guest and admin access.

## CERTIFICATIONS

Microsoft AI & ML Engineering Professional Certificate | Microsoft
IBM Machine Learning | IBM
AI and Machine Learning Algorithms and Techniques | Microsoft
Foundations of AI and Machine Learning | Microsoft
AWS Educate Machine Learning Foundations | Amazon Web Services
Mathematics for Machine Learning: Linear Algebra | Imperial College London
Data Structures | University of California San Diego
Database Management Essentials | University of Colorado System
AI For Everyone | DeepLearning.AI
Introduction to Generative AI and Prompt Engineering | Mapua University
Interactivity with JavaScript | University of Michigan
Python Classes and Inheritance | University of Michigan
Programming for Everybody (Getting Started with Python) | University of Michigan
Technical Support Fundamentals | Google
Introduction to Cybersecurity | Cisco
Welcome to Game Theory | The University of Tokyo